UNIT-I

BASIC MATHEMATICAL & STATISTICAL TECHNIQUES

Statistics is nothing but approximate value.

Data → Analysis → Conclusion

Data
- ungrouped / Individual → x Values
- Grouped
  - Discrete / finite →

x	0	1	2	3
f	4	5	6	4

  - continuous →

C.I x	0-5	5-10	10-15	15-20
f	10	15	5	25

Measures of central Tendency

A single value represents the whole value is called central Tendency.

* Arithmetic Mean (Average)
* Median (Middle Value)
* Mode (Repeated Value)
* Geometric Mean
* Harmonic Mean

Arithmetic Mean or Average

The arithmetic mean is denoted by $\bar{x}$.

$$\bar{x} = \frac{\text{Sum of observations}}{\text{No. of observations}}$$

Individual Data:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Discrete Data:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i \cdot f_i}{\sum_{i=1}^{n} f_i}$$

$$\boxed{\Sigma f_i = N}$$

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Continuous Data

Direct Method:

$$\bar{x} = \frac{\sum x_i \cdot f_i}{\sum f_i}$$

Indirect Method:

$$\bar{x} = A + \left[\frac{\sum f_i \cdot d_i}{\sum f_i} \times h \right]$$

where $d_i = \frac{x_i - A}{h}$ or $\frac{m - A}{h}$

m - indicates middle value of the C.I.

A - Any Arbitrary value in mid values.

h - width of the class interval / difference.

Calculate Arithmetic Mean for the following data

2, 4, 5, 7, 3, 9, 8.

Arithmetic Mean $(\bar{x}) = \frac{\sum x_i}{n}$

$$= \frac{2+4+5+7+3+9+8}{7}$$

$$= \frac{38}{7}$$

$$= 5.43$$

Calculate A.M. for the following data

x	2	4	6	8
f	9	8	5	4

A.M. $(\bar{x}) = \frac{\sum x_i \cdot f_i}{\sum f_i}$

$$= \frac{112}{26}$$

$$= 4.3076$$

x	f	$x_i \cdot f_i$
2	9	18
4	8	32
6	5	30
8	4	32
	26	112

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Continuous Data:

Find the mean for the following data

x	0-10	10-20	20-30	30-40	40-50
f	15	12	11	10	13

Direct Method:

$$\bar{x} = \frac{\sum x_i \cdot f_i}{\sum f_i}$$

$$= \frac{1465}{61}$$

$$= 24.016$$

x	f_i	m_i	$m_i f_i$
0-10	15	5	75
10-20	12	15	180
20-30	11	25	275
30-40	10	35	350
40-50	13	45	585
	61		1465

Indirect Method:

$$\bar{x} = A + \left(\frac{\sum f_i d_i}{\sum f_i} \times h\right)$$

$$= 25 + \left(\frac{-6}{61} \times 10\right)$$

$$= 25 - 0.9836$$

$$= 24.0164$$

x	f	m	$d = \frac{m-A}{10}$	$f_i \cdot d_i$
0-10	15	5	-2	-30
10-20	12	15	-1	-12
20-30	11	(25) A	0	0
30-40	10	35	1	10
40-50	13	45	2	26
	61			-6

Find the mean for the following data.

x	0-10	10-20	20-30	30-40	40-50
f	75	50	84	69	59

Direct Method:

$$\bar{x} = \frac{\sum x_i \cdot f_i}{\sum f_i}$$

$$= \frac{8295}{337}$$

$$= 24.6142$$

x	f_i	m_i	$m_i f_i$
0-10	75	05	375
10-20	50	15	750
20-30	84	25	2100
30-40	69	35	2415
40-50	59	45	2655
	337		8295

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Indirect Method:

$$\bar{x} = A + \left(\frac{\Sigma f_i d_i}{\Sigma f_i} \times h\right)$$

$$= 25 + \left(\frac{-13}{337} \times 10\right)$$

$$= 25 - 0.3857$$

$$= 24.6142$$

x	f	m	$d = \frac{m-A}{h}$	$f_i d_i$
0-10	75	5	-2	-150
10-20	50	15	-1	-50
20-30	84	(25) A	0	0
30-40	69	35	1	69
40-50	59	45	2	118
	337			-13

Median:

The median is also called positional average. The median is middle value of the given data.

Individual Data / ungrouped Data:

Find median for the following data 4,3,7,2,1,9,8.

first we arrange the given data in ascending order or descending order.

Ascending order: 1, 2, 3, 4, 7, 8, 9.

No. of Observations $n = 7$ Odd

$$\left(\frac{n+1}{2}\right)^{th} \text{ item} = \left(\frac{7+1}{2}\right)^{th} = 4^{th} \text{ item}$$

$\therefore$ Median = 4.

Find median for the following data 4,3,7,2,1,9,8,10.

Ascending order: 1, 2, 3, 4, 7, 8, 9, 10.

No. of Observations $n = 8$ even

If n is even $= \left[\frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2}+1\right)^{th}}{2}\right]$

$$= \left[\frac{\left(\frac{8}{2}\right)^{th} + (4+1)^{th}}{2}\right]$$

$$= \frac{11}{2}$$

$$= 5.5$$

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Calculate median for the following data

5·5, 1·5, 2·5, 0·5, 3·5, 3·6, 1·7, 0·9, 0·05, 1·05

Ascending order: 0·05, 0·5, 0·9, 1·05, 1·5, 1·7, 2·5, 3·5, 3·6, 5·5

No. of observations $n = 10$

If n is even $= \left[\frac{\left(\frac{n}{2}\right)^{th} + \left(\frac{n}{2}+1\right)^{th}}{2}\right]$

$= \frac{5^{th} + 6^{th}}{2}$

$= \frac{1.5 + 1.7}{2}$

$= 1.6$

Discrete Data:

In discrete data, the process for median

1. Find $N = \Sigma f_i$.
2. Calculate $\frac{N}{2}$.
3. Calculate cumulative frequency.
4. Choose the cumulative frequency which is just greater than $N/2$.
5. The corresponding value of x which is just greater than $N/2$ is the median.

Find the median for the following data

x	2	4	6	8	10	12	14	16
f	5	7	11	4	10	12	9	8

x	f	C.F
2	5	5
4	7	12
6	11	23
8	4	27
10	10	37
12	12	49
14	9	58
16	8	66

$\frac{N}{2} = \frac{66}{2} = 33$

Here $37 > \frac{N}{2}$

The value of x just greater than $\frac{N}{2}$ is 10.

$\therefore$ Median = 10.

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Find the median for the following data

x	10	15	16	17	20
f	35	25	45	15	55

x	f	C.F.
10	35	35
15	25	60
16	45	105
17	15	120
20	55	175
	175	

$\frac{N}{2} = \frac{175}{2} = 87.5$

Here 105 just $> \frac{N}{2}$

The value of x just $> \frac{N}{2}$ is 16.

∴ Median = 16.

Median in Continuous Data:

$$\text{Median} = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

where l = lower limit of the median class.

h = width of the interval / Interval difference

f = frequency of the median class.

N = Total frequency Σf_i

c = preceeding cumulative frequency of the median class.

Calculate median for the following data

x	0-5	5-10	10-15	15-20	20-25	25-30
f	5	7	11	9	12	8

we know that median $= l + \frac{h}{f}\left(\frac{N}{2} - c\right)$

Here first we find median class.

Median class is the class interval corresponding to the cumulative frequency which is just $> \frac{N}{2}$.

$\frac{N}{2} = \frac{52}{2} = 26.$

32 just $> 26.$

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x	f	C.F.
0-5	5	5
5-10	7	12
10-15	11	23 C
15-20	9 f	32
20-25	12	44
25-30	8	52
	52	

Here 15-20 is the median class

$$\text{Median} = 15 + \frac{5}{9}\left(\frac{52}{2} - 23\right)$$

$$= 15 + \frac{5}{9}(26-23)$$

$$= 15 + 1.66$$

$$= 16.66$$

Calculate median for the following data

x	0-5	5-10	10-15	15-20	20-25	25-30
f	15	17	12	19	16	18

we know that median $= l + \frac{h}{f}\left(\frac{N}{2} - C\right)$

Here first we find median class.

Median class is the class interval corresponding to the cumulative frequency which is just $> \frac{N}{2}$.

x	f	C.F.
0-5	15	15
5-10	17	32
10-15	12	44
15-20	19	63
20-25	16	79
25-30	18	97
	97	

$$\frac{N}{2} = \frac{97}{2} = 48.5$$

63 just > 48.5

Here 15-20 is the median class

$$\text{Median} = 15 + \frac{5}{19}\left(\frac{97}{2} - 44\right)$$

$$= 15 + \frac{5}{19}(48.5 - 44)$$

$$= 15 + 1.1842$$

$$= 16.1842$$

Mode:

Mode is the most repeated value in the given data.

Individual Data:

Find mode in the following data. 2,2,3,3,2,5,7,2,3,7,5,2

$\therefore$ Mode = 2.

Find mode in the following data. 2,2,3,3,2,5,7,2,3,7,5,2,3,3.

$\therefore$ Mode = 2, 3 (Bi-modal)

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Discrete Data:

The mode is the value which is corresponding to highest frequency.

x	2	4	6	8	10
f	10	12	13	14	9

In the given data, 8 repeated 14 times.

$\therefore$ Mode = 8.

Continuous Data:

$$\text{Mode} = l + \left(\frac{h\,(f_1 - f_0)}{2f_1 - f_0 - f_2}\right)$$

h = width of the class interval

l = lower limit of the model class

f_1 = frequency of the model class

f_0 = preceeding frequency of the model class

f_2 = suceeding frequency of the model class

Find mode for the following data

x	0-10	10-20	20-30	30-40	40-50
f	11	12	15	14	9

we know that $\text{Mode} = l + \left(\frac{h\,(f_1 - f_0)}{2f_1 - f_0 - f_2}\right)$

Here the model class is 20-30.

$l = 20,\ h = 10,\ f_1 = 15,\ f_0 = 12,\ f_2 = 14.$

$$\text{Mode} = 20 + \left(\frac{10\,(15-12)}{2(15) - 12 - 14}\right)$$

$$= 20 + \left(\frac{10\,(3)}{30 - 26}\right)$$

$$= 20 + \left(\frac{30}{4}\right)$$

$$= 20 + 7.5$$

$$= 27.5$$

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Find mode for the following data

x	0-5	5-10	10-15	15-20	20-25	25-30
f	22	21	25	27	32	31

we know that mode $= l + \left(\frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}\right)$

Here the model class is 20-25

$l = 20, h = 5, f_1 = 32, f_0 = 27, f_2 = 31.$

$$\text{Mode} = 20 + \left(\frac{5(32-27)}{2(32) - 27 - 31}\right)$$

$$= 20 + \left(\frac{5(5)}{64 - 58}\right)$$

$$= 20 + \left(\frac{25}{6}\right)$$

$$= 20 + 4.166$$

$$= 24.166$$

Geometric Mean:

Let a and b are any two numbers then the

$$G.M.\ (G_1) = \sqrt{ab} = (ab)^{1/2}$$

For 'n' numbers, let $x_1, x_2, x_3 \cdots x_n$ are n numbers.

$$G_1 = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdots x_n}$$

$$G_1 = (x_1 \cdot x_2 \cdot x_3 \cdots x_n)^{1/n}$$

For the simplification apply log on both sides

$$\log G_1 = \log (x_1 \cdot x_2 \cdots x_n)^{1/n} \qquad \log a^b = b \log a$$

$$= \frac{1}{n} \log (x_1 \cdot x_2 \cdots x_n) \qquad \log ab = \log a + \log b$$

$$= \frac{1}{n} (\log x_1 + \log x_2 + \cdots + \log x_n)$$

$$\log G_1 = \frac{1}{n} \sum_{i=1}^{n} \log x_i$$

$$\boxed{G_1 = \text{Antilog}\ \frac{1}{n} \sum_{i=1}^{n} \log x_i}$$ Individual Data

Grouped Data: (Discrete, Continuous)

$$G_1 = \text{Anti log}\ \frac{1}{N} \sum_{i=1}^{n} f_i \log x_i$$

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Find G.M. for the following data 25, 40.

$$G_1 = \sqrt{ab}$$

$$= \sqrt{25 \times 40}$$

$$= \sqrt{1000} = 31.622.$$

Find G.M. for the following data 2, 7, 5, 9, 11.

$$G_1 = \sqrt[n]{x_1 \cdot x_2 \cdots x_n}$$

$$= \sqrt[5]{2 \times 7 \times 5 \times 9 \times 11}$$

$$= (6930)^{1/5} = 6930 \wedge (1 \div 5)$$

$$= 5.8633$$

Find G.M. for the following data

x	2	4	6	8
f	5	9	7	11

$$G_1 = \text{Antilog } \frac{1}{N} \sum_{i=1}^{n} f_i \cdot \log x_i$$

$$= \text{Antilog } \left[\frac{1}{32} (51.214)\right]$$

$$= \text{Antilog } (1.6)$$

$$= 39.8107 \quad 4.9530$$

x	f	$\log x_i$	$f \cdot \log x_i$
2	5	$\ln 2 = 0.69$	3.45
4	9	$\ln 4 = 1.38$	12.42
6	7	$\ln 6 = 1.79$	12.53
8	11	$\ln 8 = 2.074$	22.814
	32		51.214

Antilog → shift + log base 10

shift + ln base e

Find G.M. for the following data

x	0-10	10-20	20-30	30-40
f	5	9	7	11

x	f	M	$\log x_i$	$f \cdot \log x_i$
0-10	5	5	1.60	8
10-20	9	15	2.70	24.3
20-30	7	25	3.21	22.47
30-40	11	35	3.55	39.05
	32			93.82

$$G_1 = \text{Antilog } \frac{1}{N} \Sigma f_i \cdot \log x_i$$

$$= \text{Antilog } \left[\frac{1}{32} (93.82)\right]$$

$$= \text{Antilog } (2.9318)$$

$$= 854.6730$$

$$= 18.7614$$

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Harmonic Mean:

Reciprocal of A.M. of the reciprocals.

Individual Data

$$H.M. = \frac{1}{\frac{1}{n}\sum\frac{1}{x_i}}$$

Grouped Data

$$H.M. = \frac{1}{\frac{1}{N}\sum f_i \cdot \frac{1}{x_i}}$$

Calculate H.M. for the following data 4, 9, 5, 7, 12.

we know that $H.M. = \frac{1}{\frac{1}{n}\sum\frac{1}{x_i}}$

$$= \frac{1}{\frac{1}{5}\left[\frac{1}{4}+\frac{1}{9}+\frac{1}{5}+\frac{1}{7}+\frac{1}{12}\right]}$$

$$= 0.135.$$

Calculate H.M. for the following data

x	5	4	3	2	1
f	7	11	13	8	12

x	f_i	$\frac{1}{x_i}$	$f_i \cdot \frac{1}{x_i}$
5	7	0.2	1.4
4	11	0.25	2.75
3	13	0.33	4.29
2	8	0.50	4.00
1	12	1.00	12
	51		24.44

we know that

$$H.M. = \frac{1}{\frac{1}{N}\sum f_i \cdot \frac{1}{x_i}}$$

$$= \frac{51}{24.44} = 2.08674$$

Calculate H.M. for the following data

x	5-10	10-15	15-20	20-25	25-30
f	21	24	27	22	–

x	f_i	M	$1/x_i$	$f_i \cdot \frac{1}{x_i}$
5-10	21	7.5	0.13	2.73
10-15	24	12.5	0.08	1.92
15-20	27	17.5	0.057	1.539
20-25	22	22.5	0.044	0.968
	94			7.157

we know that

$$H.M. = \frac{1}{\frac{1}{N}\sum f_i \cdot \frac{1}{x_i}}$$

$$= \frac{1}{\frac{1}{94}(7.157)}$$

$$= \frac{1}{0.07613}$$

$$= 13.1354$$

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Measures of Dispersion

* Range
* Quartile Deviation
* Mean Deviation
* Standard Deviation
* Coefficient of Dispersion
* Coefficient of Variation.

ungrouped Data:

* Range = Maximum - Minimum
* Quartile Deviation = $\frac{Q_3 - Q_1}{2}$

$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item $\quad Q_3 = 3\left(\frac{n+1}{4}\right)^{th}$ item

* Mean Deviation = $\frac{\sum |x_i - \bar{x}|}{n}$
* Standard Deviation $(\sigma) = \sqrt{\text{Variance}} = \sqrt{\sigma^2}$

$$\text{Variance } (\sigma^2) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$\text{S.D. } (\sigma) = \sqrt{\frac{1}{n}\sum (x_i - \bar{x})^2}$$

* Coefficient of Dispersion = $\frac{\sigma}{\bar{x}} = \frac{\text{Standard deviation}}{\text{Mean}}$
* Coefficient of Variation = $\frac{\sigma}{\bar{x}} \times 100$

Calculate measures of dispersion for the following data 4, 7, 3, 9, 12, 10, 15.

* Range = Maximum - Minimum
  = 15 - 3 = 12
* Quartile Deviation = $\frac{Q_3 - Q_1}{2}$

$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item $\quad Q_3 = 3\left(\frac{n+1}{4}\right)^{th}$ item

3, 4, 7, 9, 10, 12, 15 $\quad n = 7$, Odd $\quad = 3(2)^{th}$ item

$Q_1 = \left(\frac{7+1}{4}\right)^{th}$ item $= \left(\frac{8}{4}\right)^{th}$ item $\quad = 6^{th}$ item

$= 2^{nd}$ item $= 4$ $\quad = 12$.

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Quartile Deviation $= \frac{12-4}{2} = 4$.

3, 4, 7, 9, 10, 12, 15, 17 $\quad n=8$, Even

$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item

$= \left(\frac{8+1}{4}\right)^{th}$ item $= 2.25^{th}$ item $= 2^{nd}$ item $+ 0.25\,(3^{rd}$ item $- 2^{nd}$ item$)$

$= 4 + 0.25\,(7-4) = 4 + 0.75 = 4.75$

$Q_3 = 3\left(\frac{n+1}{4}\right)^{th}$ item $= 3\left(\frac{9}{4}\right) = 6.75^{th}$ item

$Q_3 = 6^{th}$ item $+ 0.75\,(7^{th}$ item $- 6^{th}$ item$)$

$= 12 + 0.75\,(15-12) = 12 + 0.75\,(15-12)$

$= 12 + 0.75(3) = 12 + 2.25 = 14.25$

$Q.D = \frac{14.25 - 4.75}{2} = \frac{9.5}{2} = 4.75$

* Mean Deviation $= \frac{\Sigma |x_i - \bar{x}|}{n}$

$\bar{x} = \frac{\Sigma x_i}{n} = \frac{60}{7} = 8.57$

x	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$
3	-5.57	5.57
4	-4.57	4.57
7	-1.57	1.57
9	0.43	0.43
10	1.43	1.43
12	3.43	3.43
15	6.43	6.43
	0.01	23.43

$M.D. = \frac{1}{7}(23.43)$

$= 3.3471$

* $C.D. = \frac{\sigma}{\bar{x}} = \frac{3.9589}{8.57} = 0.4619$

* $C.V. = \frac{\sigma}{\bar{x}} \times 100$

$= 46.19$

* Standard Deviation $(\sigma) = \sqrt{\text{Variance}} = \sqrt{\sigma^2}$

$\sigma^2 = \frac{1}{n}\Sigma (x_i - \bar{x})^2$

x	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
3	-5.57	31.0249
4	-4.57	20.8849
7	-1.57	2.4649
9	0.43	0.1849
10	1.43	2.0449
12	3.43	11.7649
15	6.43	41.3449
		109.7143

$\sigma^2 = \frac{1}{7}(109.7143)$

$= 15.673$

$\sigma = \sqrt{15.673}$

$= 3.9589$

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Grouped Data:

* Range = Maximum - Minimum
* Quartile Deviation = $\frac{Q_3 - Q_1}{2}$
* Mean Deviation = $\frac{1}{N} \sum f_i |x_i - \bar{x}|$
* Standard Deviation (σ) = $\sqrt{\text{Variance}} = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$
* Coefficient of Dispersion = $\frac{\sigma}{\bar{x}}$
* Coefficient of Variation = $\frac{\sigma}{\bar{x}} \times 100$

Calculate measures of dispersion for the following data

x	2	4	6	8	10
f	5	11	9	7	12

* Range = Maximum - Minimum

= 10 - 2 = 8

* Quartile Deviation = $\frac{Q_3 - Q_1}{2}$

$N = 44$, $\frac{N}{2} = \frac{44}{2} = 22$.

$Q_1 = \frac{1}{4} \cdot N = 11$. $Q_3 = \frac{3}{4} \cdot 44 = 33$.

Quartile Deviation = $\frac{10-4}{2} = 3$.

Here we calculate Q_1, Q_3 like median in discrete case.

x	f	C.F.
2	5	5
Q_1 4	11	16 > 11
Q_2 6	9	25
Q_3 8	7	32
↳ 10	12	44 > 33
	44	

* M.D. = $\frac{1}{N} \sum f_i |x_i - \bar{x}|$

$\bar{x} = \frac{\sum x_i f_i}{\sum f_i \text{ or } N}$

$= \frac{284}{44}$

$= 6.45$

x_i	f_i	$x_i f_i$	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$	$f_i \lvert x_i - \bar{x} \rvert$
2	5	10	-4.45	4.45	22.25
4	11	44	-2.45	2.45	26.95
6	9	54	-0.45	0.45	4.05
8	7	56	1.55	1.55	10.85
10	12	120	3.55	3.55	42.6
	44	284			106.7

M.D. = $\frac{1}{44}$ (106.7)

= 2.425

* C.D = $\frac{\sigma}{\bar{x}}$

= $\frac{2.7588}{6.45} = 0.4277$

* C.V. = $\frac{\sigma}{\bar{x}} \times 100$

= 0.4277 × 100

= 42.77

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* S.D. (σ) = $\sqrt{\text{Variance}}$

$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$

$= \frac{1}{44} (334.91)$

$= 7.6115$

$\sigma = 2.7588$

x_i	f_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
2	5	-4.45	19.8025	99.0125
4	11	-2.45	6.0025	66.0275
6	9	-0.45	0.2025	1.8225
8	7	1.55	2.4025	16.8175
10	12	3.55	12.6025	151.23
	44	0	41.0125	334.91

Continuous Data:

Calculate Measures of dispersion for the following Data.

x	0-10	10-20	20-30	30-40	40-50
f	12	10	11	15	13

* Range = Maximum - Minimum

= 50 - 0 = 50.

* Quartile Deviation = $\frac{Q_3 - Q_1}{2}$

Median = $l + \frac{h}{f}\left(\frac{N}{2} - c\right)$ $\frac{N}{2} = \frac{61}{2} = 30.5$

$Q_1 = l + \frac{h}{f}\left(\frac{N}{4} - c\right)$ $\frac{N}{4} = 15.25$

$= 10 + \frac{10}{10}(15.25 - 12)$ $\frac{3N}{4} = 45.75$

$= 13.25$

$Q_3 = l + \frac{h}{f}\left(\frac{3N}{4} - c\right)$

$= 30 + \frac{10}{15}(45.75 - 33) = 38.5$

x	f	C.F.
0-10	12	12
10-20	10	22 > $\frac{N}{4}$
20-30	11	33 > $\frac{N}{2}$
30-40	15	48 > $\frac{3N}{4}$
40-50	13	61
	61	

Q.D = $\frac{38.5 - 13.25}{2}$

= 12.625

* Mean Deviation = $\frac{1}{N} \sum f_i |x_i - \bar{x}|$

$\bar{x} = \frac{1595}{61}$

$= 26.14$

M.D = $\frac{1}{61}(755.7)$

$= 12.388$

x	x_i	f_i	$x_i \cdot f_i$	$x_i - \bar{x}$	$\lvert x_i - \bar{x} \rvert$	$f_i \lvert x_i - \bar{x} \rvert$
0-10	5	12	60	-21.14	21.14	253.68
10-20	15	10	150	-11.14	11.14	111.4
20-30	25	11	275	-1.14	1.14	12.54
30-40	35	15	525	8.86	8.86	132.9
40-50	45	13	585	18.86	18.86	245.18
		61	1595			755.7

* S.D. (σ) = $\sqrt{\text{Variance}}$

$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$

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$\sigma^2 = \frac{1}{61}(12419.639)$

$= 203.6006$

$\sigma = \sqrt{203.6006}$

$= 14.2688$

x	x_i	f_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
0-10	5	12	-21.14	446.899	5362.788
10-20	15	10	-11.14	124.099	1240.99
20-30	25	11	-1.14	1.299	14.289
30-40	35	15	8.86	78.499	1177.485
40-50	45	13	18.86	355.699	4624.087
		61		1006.495	12419.639

Coefficient of Deviation $= \frac{\sigma}{\bar{x}}$

$= \frac{14.2688}{26.14}$

$= 0.5458$

Coefficient of Variance $= \frac{\sigma}{\bar{x}} \times 100$

$= 54.58$

Correlation:

The relation between any two variables is called correlation (or) The change in one variable effects the change in another variable.

Example: $y = a + bx$, Here y is depending on x.

x is independent variable, y is dependent variable.

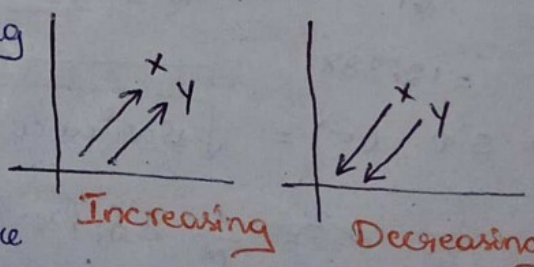
Types of Correlation:

The correlation is two types. They are positive correlation, Negative correlation.

positive Correlation:

The two variables x & Y are deviating in same direction (Increasing or Decreasing) is called positive correlation.

Example: Income & Expenditure



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